P-P-00105a (GSA-FSS)
August 22, 1962

Interim Revision of
P-P-105
December 11, 1952

INTERIM FEDERAL SPECIFICATION

PAPER, ABRASIVE (FLINT)

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized Federal Agencies to use this Interim Federal Specification as a valid exception to Federal Specification P-P-105 dated December 11, 1952.

1. SCOPE AND CLASSIFICATION

1.1 Scope. - Flint paper sheets and rolls, covered by this specification is a paper coated on one surface with a flint abrasive.

1.2 Classification

1.2.1 Classes. - Flint paper shall be of the following classes, as specified (see 6.2).

Class 1. - Sheets, flint finishing paper (see 3.4.1)
Class 2. - Sheets, flint paper (see 3.4.1)
Class 3. - Rolls, flint paper (see 3.4.2)

1.2.2 Grit size. - Flint paper shall be furnished in the above classes and grit sizes, as specified (see 6.2):

Class 1: 7/0, 6/0, 5/0, 4/0, 3/0, 2/0
Class 2: Extra Fine, Fine, Medium, Coarse, Extra Coarse
Class 3: Extra Fine, Fine, Medium, Coarse

1.2.3 Dimensions of sheet and rolls (see 3.4).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

FSC 5350

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2.1 Specifications and standards. - The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specification

UU-P-31 - Paper, General Specifications and Methods of Testing.

Federal Standards

Fed. Std. No. 102 - Preservation, Packaging, and Packing Levels
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.)

Military Specification

MIL-P-3816 - Packaging and Packing of Abrasives and Abrasive Products for Shipment and Storage.

Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Department of Commerce Commercial Standard:

CS-217 - Grading of Abrasive Grain on Coated Abrasive Products

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material

3.1.1 Abrasive

3.1.1.1 Flint. - The abrasive shall be crushed or ground from clean hard quartz, quartzite, or flint and shall be relatively free from extraneous materials.

3.1.1.2 Grading, control and tolerance. - Grading of abrasive particles shall be in accordance with Commercial Standard CS-217; and the tolerance therein shall apply to the mineral recovered from the coated abrasive product.

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3.1.2 Backing. - The backing shall be a good quality paper suitable for the purpose. It shall meet the requirements of Tables I and II directly after being conditioned at 70 ± 2°F and 50 ± 2 percent relative humidity for at least 24 hours.

Table I. - Weight of Paper Backing

Class No.	Grit Size	Nominal Paper Weight 24x36 Inches 480 Sheets	Tolerance
		Pounds	Pounds
1	7/0 through 2/0	40	37.5 - 42.5
2	Extra fine, fine,	50 to 60	46.5 - 63.0
2	Medium, coarse, extra coarse	55 to 60	51.0 - 63.0
3	Extra fine through coarse	130	122.0 - 135.0

3.1.3 Adhesive. - The adhesive shall be a good quality, animal hide glue base, suitable for the purpose intended.

3.2 Coating

3.2.1 Class 1. - The flint finishing paper sheets shall be open coat. The abrasive flint grain shall be applied to one side of the paper backing only, and cover approximately 50 percent of the backing, thus leaving open spaces between the grains of abrasive (see 6.6).

3.2.2 Class 2 and 3. - The flint paper sheets and rolls shall be closed coat. The abrasive flint grain shall be applied to one side of the paper backing only, and completely cover the surface of the backing (see 6.6)

3.3 Tensile breaking strength. - The average breaking strength of coated abrasive paper shall be in accordance with Table II when tested as described in 4.4.2.

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Table II - Tensile Breaking Strengths

Class No.	Grit Size	Tensile Strength, Average	
		Machine Direction	Across Machine Direction
		Pounds/Inch	Pounds/Inch
1	7/0 through 2/0	20	10
2	Extra Fine, fine	35	15
	Medium, coarse, extra coarse	40	20
3	Extra fine through coarse	85	45

3.4 Form and dimensions.

3.4.1 Sheet. - Unless otherwise specified (see 6.2), flint paper shall be furnished in the following sizes with a tolerance of ± 1/8 inch in length and of ± 1/16 inch in width, when measured at, and after conditioning at 70 ± 2°F and 50 ± 2 percent relative humidity for at least 24 hours:

Class 1 - 9 x 11 inches
Class 2 - 9 x 10 inches

3.4.2 Roll. - Class 3 flint paper shall be furnished in rolls not less than 50 yards long, and in the widths as specified (see 6.2).

3.5 Identification marking.

3.5.1 Class 1. - Each sheet of class 1 finishing paper shall be marked in a legible manner with the grade or grit designation.

3.5.2 Class 2. - Each sheet shall be marked in a legible permanent manner with the following information:

(a) Manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

(b) The name of the abrasive or the abrasive manufacturer's trade designation of such known character for the abrasive that the abrasive grain may be readily determined.

(c) The grade or grit designation representing the grain size used.

3.5.3 Class 3. - Each roll 6 inches and wider shall be marked at least once in every lineal yard with the grade designation.

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3.6 Workmanship. - Abrasive flint paper shall be free from imperfections affecting its appearance or which may impair its serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. - The sample size levels of inspection shall be in accordance with Military Standard MIL-STD-105.

4.2.1 Lot. - All abrasive paper of the same class, and grit size manufactured under essentially the same conditions and offered for delivery at one time shall be considered a lot for the purposes of examinations and tests.

4.2.2 Sampling for examination of packaging and packing. - Packaged and packed abrasive paper in shipping containers shall be sampled in accordance with inspection level I.

4.2.3 Sampling for visual, dimensional, and workmanship characteristics of end item. - Sampling for these characteristics shall be in accordance with inspection level L7.

4.2.4 Sampling for tests. - Samples for test shall be selected in accordance with inspection level L3. Samples may be selected from those selected in accordance with 4.2.3.

4.3 Examination and tests. - The acceptable quality levels (AQL's) shall be in accordance with Military Standard MIL-STD-105.

4.3.1 Examination

4.3.1.1 Packaged and packed abrasive paper. - Abrasive paper, ready for shipment, selected in accordance with 4.2.2, shall be examined with AQL's and classification of defects in accordance with Military Specification MIL-P-3816.

4.3.1.2 Individual sheets and rolls. - Samples of abrasive paper selected in accordance with 4.2.3 shall be examined in accordance with table III. The AQL's for defects listed in table III shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.2 Lot acceptance tests. - Samples selected in accordance with 4.2.4 shall be subjected to the tests in 4.4, AQL 6.5 percent defective.

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Table III - Classification of Defects, End Item

Examine Applicable Paragraph	Defects	Major	Minor
1.2.2	Size of grit not as specified	x	
3.1.1.1	Extraneous material present in abrasive	x	
3.2	Abrasive not closed or open coat as specified	x	
3.2	Coating not completely covering surface of backing	x	
3.4.1	Dimensions of sheets, not correct size or not within tolerances	x	
3.4.2	Size of rolls not as specified	x	
3.5	Identification marking, omitted, illegible, incomplete, or incorrect		x
3.6	Sheets and rolls, torn or cracked	x	

4.4 Tests

4.4.1 Grain Size

4.4.1.1 Grading for grain size. - Grading to determine the grain (grit) size, including recovery of grain from the sheets, preparation of grain for grading, and procedures shall be in accordance with the grading of grain as included in Commercial Standard CS-217.

4.4.1.2 Certification of grading. - Grading tests specified in 4.4.1.1 shall be omitted in instances where the contractor produces certification in writing that the abrasive grain complies with the requirements of Commercial Standard CS-217 for flint grain. Contractors are urged to include this certification with material offered for inspection and inspectors are urged to determine whether such certification is included with such material in order to preclude the necessity for grading tests.

4.4.2 Breaking strength. - The tensile breaking strength of the coated abrasive paper, directly after conditioning for at least 24 hours at 70 $\pm$ 2°F and 50 $\pm$ 2 percent relative humidity shall be determined by method 171 described in Federal Specification UU-P-31, except that the jaws of the testing machine shall be 5 inches apart, and that six specimens one inch wide shall be taken both longitudinally and transversely from sample sheets and rolls. The average tensile strength in both directions shall be computed from at least three specimens. Samples tearing at the jaws of the tensile testing machine should be discarded (see 3.3).

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5. PREPARATION FOR DELIVERY

(For civil agency procurement, the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102)

5.1 Preservation, packaging and packing. - Preservation, packaging and packing of abrasive paper shall be in accordance with Military Specification MIL-P-3816, as specified for level A, B, or C (see 6.2).

5.2 Marking

5.2.1 Civil agencies. - In addition to any special markings required by the contract or order packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

5.2.2 Military activities. - In addition to any special marking required in the contract or order packages and shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. - Abrasive flint paper can be used for all ordinary sanding of wood and painted surfaces.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and grit size for sheets (see 1.2).
(c) Class, grit size, and width of rolls (see 1.2 and 3.4.2).
(d) Level of packaging and level of packing required (see 5.1).

6.3 Transportation description. - The transportation description applicable to coated abrasive paper is as follows:

Rail - Carload minimum weight 36,000 pounds.
Motor - Truckload minimum weights 27,000, 31, 500, 36,000 pounds
Subject to Rule 34, Motor Freight Classification

6.4 Flint. - A white, translucent, somewhat crystalline, variety of natural silicon dioxide having a structure such that when crushed develops sharp edges. The hardness is approximately 6.8-7.0 (Mohs scale), or 820 Knoop scale with a specific gravity of approximately 2.6.

6.5 Coating. - A closed coat is one in which the abrasive is put on in such quantity as entirely to cover the backing. An open coat covers about 50 percent of the backing thus leaving open spaces between the grains of abrasive.

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6.6 Work value. - Coated abrasives should be stored in a cool even temperature area where the relative humidity is somewhere between 40 to 50 percent. Do not store on concrete floors. The best work value will be obtained if stored and worked in rooms having a relative humidity of approximately 50 percent.

6.7 Abrasive grain size. - The size of abrasive grain is indicated in Table IV which shows the approximate relation in size between flint and artificial abrasives.

Table IV - Comparative Grit Numbers

Silicon Carbide and Aluminum Oxide Grit Numbers	Class 1	Classes 2 and 3
320 280 240	7/0 6/0 5/0	
220 180	4/0 3/0	Extra Fine (EF)
150 120	2/0	Fine (F)
100 80 60		Medium (M)
50 40		Coarse (C)
36 30		Extra Coarse (EC)

Notice. (SEE LAST PAGE)

MILITARY INTERESTS:
Army - O C E Q Sig T
Navy - Sh Mc W
Air Force - ASD

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